MILITARY SPECIFICATION

WARHEAD, 2.75 INCH ROCKET, PRACTICE XM230, WITH OR WITHOUT FUZE LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the drawings, for one type of practice warhead designated as Warhead, 2.75 Inch Rocket, Practice, XM230 (See 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition and Special Weapons, General Specification for.
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	- Quality Assurance Terms and Definitions.
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single and Multilevel Continuous Sampling procedures and Tables for Inspection by Attributes.

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DRAWINGS

ORDNANCE CORPS

8886810 - Box, Packing, Ammunition, for Warhead 2.75 Inch Rocket, HE XM151 and Smoke M156
8886811 - Container, Ammunition, Fiber, M523 for Warhead, 2.75 Inch Rocket, HE, XM151 and
9209370 - Box, Packing Ammunition for Rocket Motor: 2.75 Inch MK40 Mod 1, MK 4, MK40 Mod 0 and Warheads M151, M156, XM230
9221050 - Warhead, 2.75 Inch Rocket, Practice XM230

PUBLICATIONS

ORDNANCE CORPS

ET-9221050 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - Material shall he in accordance with applicable drawings and specifications.

3.2 Loaded assemblies. - The loaded assemblies shall comply with all of the requirements specified on Drawing (dwg.) 9221050 and with all requirements specified in applicable specifications.

3.3 Workmanship. - Loading and assembling shall be accomplished in accordance with best practices. All parts shall be free of any foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few slight scratches not exposing base material. All required marking shall be neat and sharply defined.

3.4 First Article Inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General Quality Assurance Provisions. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of Product. - At the time the Completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced In any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substaniated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification. - Using the contractor’s written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals, all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor’s records.

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4.2 First Article Inspection

4 .2.1 Submission - The contractor shall submit a sample of twenty-four (24) loaded assemblies and six (6) unloaded assemblies to the Government for testing in accordance with the provisions of 4.2.2. All parts and materials shall be the same as will be used in production and shall be obtained from the same source of supply. Prior to submission, the sample shall be inspected by the contractor to the degree necessary to assure that it conforms to the requirements of the contract. A first article sample or portion thereof, as directed by the contracting office, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be Performed. - Assemblies and components will be submitted by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection. - If any assembly or component falls to comply with any of the applicable requirements, the initial production sample shall be rejected. The Governement reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of reflection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions for production.

4.3.1 Lot formation. - The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification or drawing not affecting safety, performance, interchangeability,

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or storage, as determined by the Government Inspector, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 or MIL-STD-1235, and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies other than the items of delivery, shall be homogenous and of a size convenient to the contractor.

4.3.2 Examination. - Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Once agreement has been reached as to which method will be used (group or individual) that method will be used for all of the CD's listed in this specification. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Metal parts assembly (see dwg. 8889406 covering a detail of dwg. 9221050)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Interior coating damaged or not continuous.	Visual	01001
Minor:	AQL 0.65 percent		
201.	Foreign matter present in interior.	Visual	01002

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4.3.2.2 Warhead, loaded, prior to assembly of cap or fuze, or fuze body (see dwg 9221050)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Charge missing	Visual	02001
102.	Filler in thread	Visual	02002
103.	Distance to filler	Gage	02003
104.	Fuze cavity threads clean (when applicable)	Visual/ Manual	02004
Minor:	AQL 1.00 percent		
201.	Compound missing from base thread	Visual	02005
202.	Compound missing from nose threads if to be shipped or stored without fuze or fuze body.	Visual	02006

4.3.2.3 Warhead, loaded (see dwg 9221050)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.0 percent		
101.	Fuze or Fuze Body torque improper	Gage	03001
102.	Base thread damaged	Visual	03002
103.	Bourrelet damaged	Visual	03003
104.	Diameter of bourrelet, max	Gage	03004
105.	Nose cone improper	Visual	03005
Minor	AQL 1.00 percent		
201.	Marking missing, misleading or unidentifiable	Visual	03006
202.	Evidence of poor workmanship (see 3.3)	Visual	03007

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4.3.2.4 Fiber container, sealed, prior to taping (see dwg. 6886811)

Categories	Defects	Method of inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 0.65 percent		
201.	Gap between body and cover in excess of 1/8 inch	Gage	04001

4.3.2.5 Fiber container, sealed (see dwg. 8886811)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Tape incomplete or badly wrinkled	Visual	05001
102.	cuts, scuffs, gouges or checks which penetrate outer layer.	Visual	05002
103.	Metal end loose or distorted	Visual/Manual	05003
Minor:	AQL 1.50 percent		
201.	Length of tear tab, min	Gage	05004
202.	Marking missing, misleading or inidentifiable	Visual	05005
203.	Contents loose	Manual	05006

4.3.2.6 Box, shipping (see dwg. 8886810)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Contents exposed	Visual	06001
102.	Board broken	Visual	06002
103.	FSC number missing or misleading	Visual	06003
104.	Hardware missing, broken or loose	Visual/Manual	06004
105.	Strapping missing, broken or loose	Visual/Manual	06005

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Minor	AQL 1.50 percent		
201.	Marking misleading or Unidentifiable.....	Visual	06006
202.	Hardware improperly engaged....	Visual	06007
203.	Car seal missing, unsealed or improperly positioned.....	Visual	06008
204.	Handle missing or. insecure.....	Visual/ Manual	06009

4.3.2.7 Box, Packing, Ammunition for Rocket Motor: 2.75 Inch MK40 Mod 1, MK4, MK40 Mod 0 and Warheads M151, M156, XM230, prior to sealing (see dwg 9209570).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Contents positioned irnproperly.	Visual	07001
102.	Contents packed not as marked on outside of box.....	Visual	07002
Minor:	AQL 0.65 percent		
201.	Foreign material in box.....	Visual	07003

4.3.2.8 Box Packing Ammunition for rocket Motor: 2.75 Inch MK40 Mod 1, MK4, MK40, Mod 0 and Warheads M151, M156, XK230 (see dwg 9209570).

Categories	Defects	Method of Inspection	Code No.
Critical:	None Defined		
Major:	AQL 0.65 percent		
101.	Box damaged to extent that contents are exposed or able to become exposed.....	Visual	08001
102.	Contents loose.....	Manual	08002
103.	Hardware broken or loose.....	Visual/Manual	08003
104.	Strapping broken or loose.....	Visual/Manual	08004
Minor:	AQL 1.5 percent		
201.	Hardware Improperly engaged.....	Visual	08007
202.	Strapping Improperly assembled....	Visual	08008
203.	Car seal missing, unsealed or mislocated	Visual	08009
204.	Marking misleading or unidentifiable.....	Visual	08010

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4.3.3 Testing

4.3.3.1 Weight of charge (see dwg. 9221050) - Major defect - Code No. 02007. - This determination shall be performed on a sample of 200 units for the first production lot. The determination shall be performed on a sample of 50 units from each subsequent lot. The average weight of 25 empty units shall be used as the basis for charge weight determination. Under sampling, if three (3) or more units fail to meet the requirement, the lot shall be rejected.

4.3.4 Inspection Equipment - Equipment Tabulation Number ET-9221050 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance. - Supply and maintenance of the equipment listed on the ET shall be In accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor’s inspection equipment. The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

5. Preparation for Delivery

5.1 Packaging. - Level A

5.1.1 Warheads. - Packaging shall be in accordance with drawing 8886811.

5.2 Packing. - Level A

5.2.l Warheads. - Packing shall be in accordance with drawing 8886810.

5.2.2 Rocket Motors and Warheads shall be packed in accordance with drawing 9209570.

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5.3 Marking. - Unit and exterior containers shall be marked in accordance with marking instructions contained in the applicable drawing.

6. NOTES

6.1 Ordering data. -Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified In Standard Mil-Std-1167.

6.2 Inspection code numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as “Contractor Design” in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.

MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated

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"MU") non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - All submitted designs shall contain a reference to the applicable five digit Code Number contained In Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND9. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

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6.4 Fuze or Fuze Body .- The practice warhead will normally be supplied with either a fuze or fuze body as designated by the procuring activity.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any pattented inventton that may in any way be related thereto.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1340-A-234

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DOVER, NEW JERSEY 07801

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 1.9 R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use, the clarification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this sheet will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity*)

SUBMITTED BY (*Printed or typed name and activity*)	DATE

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NOTICE OF INACTIVATION
FOR NEW DESIGN

INCH-POUND

MIL-W-60425(AR)
NOTICE 2
12 December 1995

MILITARY SPECIFICATION

WARHEAD, 2.75 INCH ROCKET PRACTICE, XM230 WITH OR WITHOUT FUZE LOADING, ASSEMBLING AND PACKING

This notice should be filed in front of MIL-W-60425 (AR), dated 15 May 1967

MIL-W-60425(AR), dated 15 May 1967, with Validation Notice 1, dated 10 April 1987 is inactive for new design and is no longer used, except for replacement purposes.

Preparing Activity:
Army - AR

AMSC N/A

FSC 1340

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